Class 8

Chemistry

Volume 2

Solutions

Table of Content

4.	Chemical Bonding – II	1 – 3
5.	Behaviour of Gases	4 – 11
6.	Solutions	12 – 19

4. CHEMICAL BONDING - II

WORKSHEET – 1

Level – I

1. A
2. B
3. A
4. B
5. A
6. B
7. A
8. A
9. A
10. A

Level – II

1) C	2) B	3) C	4) B	5) A	6) D	7) A
8) B	9) B	10) A	11) B	12) A	13) D	14) D
15) C	16) B	17) C	18) C	19) D	20) C	21) ABC
22) A	23) C	24) D	25) a-s; b-p; c-q; d-r		26) A, B	27) A
28) B	29) A	30) a-q, s; b-p, r; c-s; d-p				

WORKSHEET – 2

Level – I

1) D	2) A	3) A	4) A	5) D	6) A	7) A
8) A	9) A	10) D				

Level – II

1) A	2) A	3) B	4) A	5) B	6) D	7) A
8) B	9) D	10) D	11) A	12) C	13) B	14) A
15) B	16) D	17) D	18) A, B	19) A, C, D	20) D	21) B
22) A	23) C	24) A	25) D	26) a-s; b-p; c-q; d-r		27) A, C
28) A	29) a-r; b-s; c-q; d-p		30) a-r; b-p; c-s; d-q			

WORKSHEET – 3

Level – I

1) B	2) B	3) D	4) A	5) B	6) C	7) C
8) C	9) C	10) C				

Level – II

1) C	2) A	3) B	4) C	5) A	6) D	7) C
8) D	9) B	10) D	11) C	12) A	13) A	14) A
15) D	16) B	17) A	18) B	19) C	20) B, C,D	21) A, C
22) A	23) A	24) C	25) A	26) C	27) 5	
28) a-r; b-p; c-q; d-q		29) A, B	30) A	31) a-r; b-p; c-s; d-q		

WORKSHEET – 4

Level – I

1) D	2) D	3) A	4) B	5) B	6) B	7) C

Level – II

1) A	2) C	3) B	4) A	5) B	6) D	7) B
8) A	9) B	10) B	11) D	12) B	13) B	14)C
15) D	16) B	17) C	18) B	19) B,D	20) C	21) A
22) C	23) C	24) A	25) a-r; b-p; c-q; d-s		26) A	27) B
28) a-q; b-p; c-r; d-s		29) B	30) A, B, C, D		31) a-q, b-r c-p, d-s	

WORKSHEET – 5

Level – I

1. C	3. A	5. D	7. B
2. A	4. D	6. A	8. C

Level – II

1) A	2) B	3) A	4) A	5) B	6) B	7) C
8) D	9) A	10) D	11) C	12) A	13) D	14) C
15) B	16) B	17) C	18) B	19) D		

WORKSHEET – 6

Level – I

1) C	2) C	3) A	4) B	5) B	6) A	7) B
8) D	9) D	10) B				

Level – II

1) C	2) C	3) B	4) A	5) A	6) B	7) B
8) A	9) A	10) B	11) D	12) D	13) C	14) A
15) A	16) C	17) D	18) A	19) B	20) D	21) A,C
22) a-r; b-p; c-s; d-q		23) B, C, D	24) A, B, C	25) D	26) A	
27) a-r; b-p; c-s; d-t		28) C	29) B	30) D	31) D	

WORKSHEET – 7

Level – I

1) C	2) A	3) A	4) A	5) B	6) C	7) A
8) D	9) C					

Level – II

1) C	2) B	3) C	4) C	5) B	6) A	7) A
8) A	9) A	10) B	11) D	12) C	13) D	14) A
15) B	16) A	17) C	18) A,B,C	19) B, D	20) A	21) C
22) A	23) A	24) C	25) a-q; b-r; c-s; d-p			

WORKSHEET – 8

Level – I

1) B	2) D	3) A	4) B	5) A	6) A	7) A
8) C						

Level – II

1) A 2) A 3) B 4) A 5) C 6) D 7) C
8) B 9) D 10) D 11) D 12) A 13) A, B, C, D
14) A, B, C 15) D 16) D 17) B 18) A 19) C
20) a-r; b-s; c-p; d-q 21) A, B, C 22) C 23) A 24) a-s; b-p; c-r; d-q
25) A, B, C, D 26) B 27) D

WORKSHEET- 9

Level – I

1) A 2) C 3) B 4) C 5) A 6) A 7) B
8) D 9) C 10) C

Level – II

1) B 2) B 3) B 4) A 5) D 6) D 7) A
8) A 9) C 10) C 11) B 12) B 13) A 14) A
15) a-s; b-r; c-p; d-q 16) D 17) D 18) C 19) A 20) 8
21) A, C 22) A 23) A, B, C 24) C 25) A 26) B
27) a-p, q; b-t; c-s; d-t; e-r, s 28) D 29) C 30) B 31) C
32) C

WORKSHEET- 10

Level – I

1) B 2) B 3) A 4) A 5) A 6) A 7) C

Level – II

1) B 2) B 3) D 4) A 5) C 6) A, D 7) B
8) B 9) A, C 10) C 11) A 12) B 13) A
14) A, B, D 15) D 16) A 17) A 18) A 19) a-p; b-q; c-s; d-r
20) 0 21) A 22) A, B, C, D 23) A, B, C 24) A 25) D
26) C 27) a-q; b-p; c-p,t; d-s 28) A, B, D 29) A, B 30) A 31) B
32) C

WORKSHEET – 11

Level – I

1) B 2) D 3) D 4) C 5) B 6) A
7) A 8) A 9) A 10) B 11) A

Level – II

1) C 2) D 3) B 4) C 5) A 6) A
7) D 8) A 9) A, B, C, D 10) B 11) A
12) C 13) A,B,D 14) A,B,C 15) D 16) D 17) B
18) a-q: b-r: c-r: d-s 19) A 20) 208.3 21) C 22) A, B
23) C 24) C 25) a-s: b-q: c-r: 0d-s 26) C 27) C
28) A,B,C,D 29) B, C

5. Behaviour of Gases

WORKSHEET – 1

Level – I

1) C	2) D	3) A	4) A	5) C	6) D	7) B
8) C	9) B	10) D				

Level – II

1) B	2) B	3) C	4) B	5) C	6) B	7) D
8) B	9) C	10) A	11) C	12) A	13) C	14) D
15) C	16) C	17) B	18) A	19) B	20) C	21) A
22) A	23) B	24) A	25) A	26) D	27) B,D	28) A,B,C
29) A	30) A	31) C	32) a-s; b-q; c-r; d-p			

1. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$;
$\frac{P_1}{P_2} = 1; \frac{V_1}{V_2} = \frac{1}{2}; \mathrm{T_1} = 273\ \mathrm{K}$,
Therefore, inserting these values into above equation,
$$\frac{P_1V_1}{P_2V_2} = \frac{T_1}{T_2} \rightarrow \frac{1}{2} = \frac{273}{T_2} \rightarrow$$
$$T_2 = 546\ \mathrm{K} = 273^{\circ}\mathrm{C}$$

2. According to Boyle's law , volume of the gas is inversely proportional to the pressure at constant temperature.
$$P_1V_1 = P_2V_2$$
Let volume be ' V ' and it becomes three times the V at $\mathrm{P_2}$.
$$PV = P_2X3V$$
$$P_2 = \frac{P}{3}$$

3. $\mathrm{PV = nRT}$

4. According to Ideal gas equation $\mathrm{PV = nRT}$, Here $\mathrm{P, V}$ and T are same. So, the number of molecules also same.
Ratio of atoms of these gases
$= H_2 : \mathrm{He} : \mathrm{O_2} : \mathrm{O_3} = 2 : 1 : 2 : 3$

5. $\mathrm{PV = nRT; P(1) = 1 \times 0.0821 \times 273}$
$\mathrm{P = 22.41\ atm}$

8. $\frac{3.5}{28} \times 6.023 \times 10^{23} = 0.75 \times 10^{23}$

9. $\frac{P_1}{P_2} = \frac{300}{600} = \frac{1}{2}$
$$n_1T_1 = n_2T_2$$
$$n \times 300 = \frac{3n}{5} \times T_2$$
$$T_2 = 500\ \mathrm{K} = 500 - 273 = 277^{\circ}\mathrm{C}$$

15. Given: At STP volume of gas in a discharge tube = 1.12×10^{-4}ml At STP, 22.4 lit volume contains 6.023×10^{23} number of molecules Number of molecules in

$$1.12 \times 10^{-4}\text{ml} = \frac{1.12 \times 10^{-4}\text{ml} \times 6.023 \times 10^{23}}{22400\text{ml}}$$

$1.12 \times 10^{-4}\text{ml} = \frac{1.12 \times 10^{-4}\text{ml} \times 6.023 \times 10^{23}}{22400\text{ml}}$

$= 3.015 \times 10^{15}$

16. From Charle' law

$V_1T_2 = V_2T_1 \rightarrow 2X299.1 = V_2X296.4$

$V_2 = 2.018lt$

17.

$d = \frac{PM}{RT} \rightarrow M = \frac{dRT}{P}$

20.

$PM = dRT$, Here,

$PV = dR300$

$PV = 0.75dRT$

From 1&2

$1 = \frac{300}{0.75T} \rightarrow T = 400K$

WORKSHEET – 2

Level – I

1)C	2) D	3) C	4) D	5) B	6) D	7) B
8) A	9) D	10) A	11) C	12) C		

Level – II

1) A	2) D	3) A	4) C	5) C	6) C	7) A
8) D	9) A	10) A	11) B	12) A	13) B	14) C
15) D	16) C	17) B	18) A	19) B	20) B	21) B
22) A	23) A,B,C	24) B,C	25) A,B,C	26) A	27) A,B,C	28) A
29) A	30) A	31) a-q; b-r; c-s; d-p		32) a-r; b-p; c-q,s; d-p		

2. $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \frac{2/2}{1/t} = \sqrt{\frac{32}{2}} \Rightarrow t = \sqrt{16} = 4hr$

3. $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{81}{100}} = \frac{9}{10}$

4. $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \frac{100}{r_2} = \sqrt{\frac{64}{16}}$

5. $\frac{r_1}{r_2} = \frac{2}{1}, \sqrt{\frac{M_2}{M_1}} = \frac{2}{1} \rightarrow \frac{M_1}{M_2} = \frac{1}{4}$

$\frac{V_1}{V_2} = \sqrt{\frac{M_2}{M_1}} = \frac{100}{V_2} = \sqrt{\frac{44}{44}} \rightarrow V_2 = 100$

7.

$\frac{V_1 t_2}{V_2 t_1} = \sqrt{\frac{M_2}{M_1}} \rightarrow \frac{180 X t_2}{120 \times 1.5}$

$= \sqrt{\frac{64}{16}} \rightarrow t_2 = 2 \text{ min}$

9.

$P_{N_2} = 760X \frac{\frac{7x}{28}}{\frac{7x}{28} + \frac{4x}{4}}$

$P_{N_2} = 760X \frac{1}{5} = 152 \text{ mm}$

$= \frac{152}{760} \text{ atm} = 0.2 \text{ atm}$

10.

The sum of all partial pressure = total pressure

$3.7 + 4.2 + CO_2 = 9.8$

Partial pressure of CO_2 is 1.9 atm

11.

Pressure due to hydrogen gas

$= \frac{1.5}{2.25} = \frac{2}{3}$

Pressure due to oxygen gas

$= \frac{0.75}{2.25} = \frac{1}{3}$

$\frac{r_A}{r_B} = \sqrt{\frac{d_B}{d_A}} \rightarrow \left(\frac{5r}{r}\right)^2 = \frac{d_B}{d_A} \rightarrow \frac{d_A}{d_B} = \frac{1}{25}$

19.

$P_A V_A + P_B V_B = P_M V_M$

$600 \times 1 + 800 \times 0.5 = P_M \times 2$

$P_M = 500 \text{ torr}$

21. $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

22. $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

WORKSHEET – 3

Level – I

1) B 2) B 3) D 4) C 5) D 6) D

Level – II

1) A 2) D 3) C 4) B 5) C 6) B 7) D

8) C 9) B 10) C 11) A 12) C,D 13) A,B,C,D 14) A

15)C 16) C 17) C 18)a-s, b-p, c-q, d-r 19) 1247 T/J

20) A,C 21) 8 22) B 23) B 24) A 25)a-r, b-p, c-s, d-q

3. Average K.E. at same temperature = 1: 1
4. $\frac{W_1}{M_1} = \frac{n}{M_2}$
5. $\frac{K \cdot E_1}{K \cdot E_2} = \frac{n_1 T_1}{n_2 T_2}$
6. $K.E = \frac{3}{2} nRT = \frac{3}{2} \times \frac{2}{2} \times 0.0821 \times 273 = 33.61$
7. $K.E = \frac{3RT}{2 \times N_o}$
8. $K.E_1 : K \cdot E_2 : K \cdot E_3 = \frac{1}{M_1} : \frac{1}{M_2} : \frac{1}{M_3}$
9. K.E. $\propto \frac{WT}{M}$
10. $\mathrm{P} = \frac{1}{3}\frac{M}{V} C^2 C = \sqrt{\frac{3RT}{M}}\ \mathrm{KE} = \frac{3}{2} RT$
11. $\mathrm{PV} = \frac{1}{3}\mathrm{mnc}^2 \Rightarrow \mathrm{C}^2 = \frac{3\mathrm{PV}}{\mathrm{M}} = \frac{3\mathrm{RT}}{\mathrm{M}}$;

    For 1 mole of gas $\mathrm{C} = \sqrt{\frac{3\mathrm{PV}}{\mathrm{mn}}} = \sqrt{\frac{3\mathrm{PV}}{\mathrm{m}}}$

    $\Rightarrow \mathrm{C} = \sqrt{\frac{3\mathrm{P}}{\left(\frac{\mathrm{m}}{\mathrm{v}}\right)}} = \sqrt{\frac{3\mathrm{P}}{\mathrm{d}}}$
14. Statement I:

    $\mathrm{KE} \propto \mathrm{T} \Rightarrow \frac{\mathrm{KE}_1}{\mathrm{KE}_2} = \frac{\mathrm{T}_1}{\mathrm{T}_2} = \frac{300}{600} = \frac{1}{2}$;

    $\therefore \mathrm{KE}_1 = \frac{1}{2}\mathrm{KE}_2 \Rightarrow \mathrm{KE}_2 = 2\mathrm{KE}_1$

    StatementII:

    $\mathrm{R} = 2\mathrm{CalT} = 273 + 27 = 300\ \mathrm{K}$

    $\Rightarrow$ Avg $\mathrm{KE} = \frac{3}{2}\mathrm{nRT}$

    $= \frac{3}{2} \times \mathrm{n} \times 2 \times 300 = 900 \mathrm{Calmol}^{-1}$

    Statement II is true
15. Given that: $\mathrm{N}_2 = 14$ g

    $\mathrm{T} = 127^{\circ}\mathrm{C} = 273 + 127 = 400$ K

    Mol mass of $\mathrm{N}_2 = 28$

    $\mathrm{R} = 8.31 \mathrm{JK}^{-1}\mathrm{Mol}^{-1} \Rightarrow$

    $\mathrm{KE} = \frac{3}{2}\mathrm{nRT} = \frac{3}{2} \times \frac{14}{28} \times 8.31 \times 400$

    $= 2493$ J
16. Given that

    $\mathrm{n}_{\mathrm{He}} = 0.3$ Mole

    $\mathrm{T}_{\mathrm{He}} = ?$

    $\mathrm{n}_{\mathrm{Ar}} = 0.4$ mole

    $\mathrm{T}_{\mathrm{Ar}} = 400$ K

    Given that $\mathrm{KE}_{\mathrm{He}} = \mathrm{KE}_{\mathrm{Ar}}$

$\Rightarrow \frac{3}{2} \times n_{He} \times R \times T_{He} = \frac{3}{2} \times n_{Ar} \times R \times T_{Ar}$

$\Rightarrow 0.3 \times T_{He} = 0.4 \times 400$

$\Rightarrow T_{He} = \frac{1600}{3} = 533\ K$

17. According to kinetic theory of gases "at constant temperature the average 22. kinetic energy of all gases is same.

$KE = \frac{3}{2} KT = \frac{3}{2} \times 1 \times 8.31\ T/J$

$[\because n = 1] = 1.5 \times 8.31\ T/J$

$= 12.465\ T/J$

$\therefore KE = 12.465\ T/J$

23. $KE = \frac{3}{2} nRT = \frac{3}{2} \times n \times 8.31 \times 298$

$= 3710.57$ molecules

KE per molecules =

$\frac{371057 \times 10^{-2}}{6.023 \times 10^{23}} = 6.16 \times 10^{-21}\ J$

24. Given that

$n = \frac{mass}{Mol.mass} = \frac{1}{32};$

$T = 47°C = 47 + 273 = 320\ K$

$KE = \frac{3}{2} nRT = 1.5 \times \frac{1}{32} \times 8.314 \times 320$

25. a)

$= 1.24 \times 10^2\ J$

$KE \propto n; \therefore n_{H_2} = \frac{8}{2} = 4$

$\Rightarrow n_{O_2} = \frac{8}{32} = \frac{1}{4};$

$\therefore \frac{KE_{H_2}}{KE_{O_2}} = \frac{n_1}{n_2} = 16$

b) Given that

$T_A = 27°C + 273 = 300\ K;$

$T_{C_2H_6} = 27 + 273 = 300\ K$

We know that $U_{rms} = \sqrt{\frac{3RT}{M}};$

$\therefore\ T \propto M \Rightarrow \frac{T_A}{T_{C_2H_6}} = \frac{M_A}{30}$

$\Rightarrow M_A = 30$ c) Given that

$n = 1;$

$T = 27 + 273 = 300\ K$

; $R = 2Cal/mole$

$KE = \frac{3}{2} nRT$

$= \frac{3}{2} \times 1 \times 2 \times 300$

$= 900Cal/mole$

d) Given that

Mass of methane = 8 g;

Mol. mass of methane = 16 g

$KE = \frac{3}{2}nRT$

$= \frac{3}{2} \times \frac{8}{16} \times 8.31 \times 300$

$= 1869.75$

WORKSHEET – 4

Level – I

1) A	2)A	3)A	4)C	5)A	6) A	7) A

Level – II

1) C	2) C	3) B	4) A	5) B	6) D	7) B
8) C	9) B	10) B	11) A	12) B	13) A,B,C,D	14) A
15) A	16) C	17) D	18) a-r,t; b-s, c-q, d-p		19) 0.9231	20) B,C
21)A,B,C,D	22) B,C,D	23) C	24) B			

2. $\frac{u_1}{u_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{0.3}{u_2} = \sqrt{\frac{300}{1200}} = \frac{1}{2}$
3. $\frac{T_1}{300} = \frac{14}{4}$
4. $\frac{u_1}{u_2} = \sqrt{\frac{M_2}{M_1}}$
5. $\frac{u_1}{u_2} = \sqrt{\frac{M_2}{M_1}}$
6. $\frac{T_1}{T_2} = \frac{M_1}{M_2}$
7. $\frac{u_1}{u_2} = \sqrt{\frac{T_1 M_2}{T_2 M_1}}$
8. $\frac{u_1}{u_2} = \sqrt{\frac{M_2}{M_1} \times \frac{T_1}{T_2}}$
9. $C_P = 0.8166 \times C$
10. $\frac{u_1}{u_2} = \sqrt{\frac{M_2}{M_1}}$
11. $u = \sqrt{\frac{u_1^2 + u_2^2 + u_3^2 + u_4^2}{4}}$
12. $U_{AV} = \sqrt{\frac{8RT}{\pi M}}$

WORKSHEET – 5

Level – I

1) C	2)C	3)A	4)D	5)D	6) B

Level – II

1) A	2) A	3) B	4) D	5) C	6) D	7) D
8) B	9) C	10) A	11) B	12) D	13) A	14) A
15) A	16) C	17) B,C,D	18) A,C	19)A,B,C	20) D	21) D
22) B	23) a-r; b-s; c-q; d-p		24) A	25) A,D	26) B	27) C
28) D	29) A,C,D	30) a-p; b-r; c-s; d-p				

2. $Z = \frac{PV_m}{RT}$

3. $Z = \frac{PV}{nRT} \therefore n = \frac{PV}{ZRT}$

$= \frac{800 \times 100}{1.95 \times 0.0821 \times 223}$

$= 2216.7$

11. $Z = \frac{PV}{nRT} = \frac{40 \times 0.4}{1 \times 0.082 \times 300} = 0.64$

12. $M = 30g; z = \frac{PV}{nRT}$

$n = \frac{100 \times 100}{0.927 \times 0.0821 \times 273}$

$\Rightarrow n = \frac{w}{30} \Rightarrow w = 14.4 \times 10^3 \text{gm}$

$= 14.4$ kg

21. $r = 1A^\circ = 10^{-10}$ m

Volume of sphere $= \frac{4}{3}\pi r^3$

$= 4.19 \times 10^{-30}$ m^3

Volume occupied by 1 mole of gas at NTP $= 4.19 \times 10^{-30}$ m$^3 \times 6.023 \times 10^{23}$

$= 2.52 \times 10^{-6}$ m^3

WORKSHEET – 6

Level – I

1) C	2) B	3) C	4)A	5) D	6) D	7) B
8) A	9) B					

Level – II

1) C	2) C	3) D	4) A	5) B	6) D	7) B
8) A	9) C	10) C	11) A	12) D	13) A	14) A
15) B	16) A	17) C	18) D	19) C	20) B	21) A
22) B	23) C	24) A,B				

1. Easily liquefiable gases have larger ' a ' value
2. The gas with higher Boyle's temperature is liquified easily
3. $b = 4 \times N_0 \times \frac{4}{3}\pi r^3$

$3.22 \times 10^{-5} = 4 \times 6.023 \times 10^{23} \times \frac{4}{3} \times \frac{22}{7} \times r^3$

$r = 0.1472 \times 10^{-9}$ m $= 0.1472$ nm

$d = 2r = 0.2944$ nm

WORKSHEET – 7

Level – I

1) A 2) B 3) C 4) B 5) A 6) B 7) A
8) B

Level – II

1) B 2) B 3) C 4) C 5) D 6) D 7) D
8) D 9) B 10) D 11) D 12) A 13) C 14) A
15) D 16) A,B 17) C 18) D 19) C 20) B
21) a-p; b-r; c-q; d-s 22) C 23) A,B,C 24) B,C

6. Solutions

WORKSHEET – 1

Level – I

1. D
2. B
3. D
4. A
5. A
6. B
7. A
8. A
9. 2
10. 4

Level – II

1) D 2) B 3) B 4) A 5) D 6) C 7) B
8) B 9) A 10) D 11) C 12) D 13) A 14) D
15) A 16) D 17) A 18) D 19) A 20) B
21) B,C,D 22)A,B,C,D 23) D 24) B 25) D 26) a-p, b-q, c-r, d-s
27) C 28) D 29) D 30) a-p,s; b-q,r; c-q,r; d-p,s 31) 1,2
32) a-r, b-p, c-q, d-s

WORKSHEET – 2

Level – I

1. A
2. A
3. D
4. A
5. C
6. C
7. A
8. C
9. A
10. A

1. $W\% = \frac{\text{Weight of solute}}{\text{Weight of solution}} \times 100$
2. $W\% = \frac{\text{Weight of solute}}{\text{Weight of solution}} \times 100; = \frac{10}{120} \times 100 = \frac{50}{6} = 8.33$
3. $V\% = \frac{\text{Volof solute}}{\text{Volof solution}} \times 100$
4. $V\% = \frac{v}{v+V} \times 100; = \frac{4}{4+36} \times 100; = \frac{4}{40} \times 100 = 10$
5. $V\% = \frac{v}{v+V} \times 100; = \frac{10}{10+40} \times 100; = \frac{10}{50} \times 100 = 20$
6. $W\% = \frac{\text{Weight of solute}}{\text{Weight of solution}} \times 100; W\% = \frac{w}{w+W} \times 100; = \frac{10}{10+190} \times 100$
   $= \frac{10}{200} \times 100 = 5$
7. PPM
8. Mass of Solute = 15gr
   Mass of solution = 100Ml
   $d = \frac{M}{V}, M = d \times V = 0.96 \times 100; M_{solution} = 96;$ Mass $\% = \frac{15}{96} \times 100; = 15.6$
9. $W\%Na_2CO_3$ means 10 gr Na_2CO_3 dissolved In 100gr solution.
   $10\%Na_2CO_3(w/V\%)$ means 10gr of Na_2CO_3 is dissolved in 100ml of solution.

Level – II

1) D 2) B 3) B 4) B 5) B 6) D 7) D
8) A 9) A 10) C 11) D 12) C 13) A 14) B
15) A 16) 10 17) A 18) C 19) D 20) B 21) D
22) A 23) A 24) C 25) A 26) B 27) a-p;b-q;c-r;d-s

28) 10 29) A, B 30) A 31) B 32) A, B 33) A, B, C

1. Strength of solution $= \frac{\text{Mass of solute(In gr)}}{\text{Vol of solution In(lit)}}$ and
   S.O.S $= \frac{\text{Mass of solute (gr)}}{\text{Vol of solution (In ml)}} \times 1000$
2. Units of strength of solution : gram Lit^{-1}
3. Normal solution
4. $10\% = \frac{w}{w+W} \times 100; 10\% = \left(\frac{x}{x+300}\right) \times 100; 10(x+300) = x \times 100; 10x + 300 = 100x$
   $3000 = 100x - 10x; 3000 = 90x; x = \frac{3000}{90} = 33.33$
5. $W\% = \frac{w}{w+W} \times 100; 25 = \frac{w}{250} \times 100; w \times 4 = 250; w = \frac{250}{4} = 62.5$ gr of solute
   $\Rightarrow W_{solution} = W_{solute} + W_{solvent}; 250 = 62.5 + x$
   $x = 250 - 62.5 = 187.5$ gr of solvent
6. Part's per million
7. $v\% = \frac{v}{v+V} \times 100; = \frac{50}{50+500} \times 100; = \frac{50}{550} \times 100 = \frac{100}{11} = 9.09$
8. 155PPM urea means 10^6ml of water has 155gr of urea
9. $PPM = \frac{\text{Mass of solute component}}{\text{Total Mass of solution } (M_1)} \times 10^6$
   $155 = \frac{x}{100} \times 10^6 \Rightarrow 155 \times 100 = x \times 10^6 \Rightarrow x = \frac{155 \times 100}{10^6} \Rightarrow 155 \times 10^{-4}$ gr of Urea.
11. $10gm = \frac{10}{180}$ moles are present in 100cc. i.e. 0.1lt,
    Hence, 1 mole of glucose will present in $\frac{0.1 \times 182}{10} = 1.8lt$
12. $(Wt\%)_{NaOH} = 6\%$
    $(Wt)_{NaOH} = 30\%$
    Mass of water = 200gr
    Mass of water = ?
    $\frac{(Wt)_{NaoH}}{\text{Mass of water}} = \frac{(Wt\%)_{NaoH}}{\text{Mass of water}}$
    $\frac{6}{200} = \frac{30}{x}; x = 1000gr$
    The amount of water to be added
    $= 1000 - 200 = 800gr$
13. $\frac{v}{V}\% = 20$; Wt of solute = 70Ml; Solvent = ?
    $V\% = \frac{v}{v+V} \times 100$
    $20 = \frac{70}{70+V} \times 100; 20(70+V) = 7000;$
    $1400 + 20v = 7000; 20v = 7000 - 1400$
    $20\ V = 5600; V = \frac{5600}{20}; V = 280ml$
14. W% = ?
    wt of solute = 15gr
    Wt of solvent = 60gr

$$Wt\% = \frac{W}{W+W} \times 100$$

$$Wt\% = \frac{15}{15+60} \times 100; Wt\% = \frac{15}{75} \times 100 = 20$$

15. Mass of glucose $= 180 \times \frac{18}{100} = 32.4gr$

16. Mass Of Glucose $= 180 - 32.4 = 147.6$ gr of H_2O

Mass of solute $= 10gr$

Mass of solvent $= 90gr$

$$\text{Mass present} = \frac{\text{wt of solute}}{\text{wt of solution}} \times 100; W\% = \frac{10}{90+10} \times 100; W\% = \frac{10}{100} \times 100 = 10$$

17. wt of solute $= 20gr$

wt of solvent $= 80gr$

$$w\% = ?; W\% = \frac{w}{w+W} \times 100; W\% = \frac{20}{20+80} \times 100; W\% = \frac{20}{100} \times 100 = 20$$

18. wt of solute $= 15gr$

wt of solution $= 150gr$

$$w\% = ?;\ W\% = \frac{w}{w+W} \times 100; W\% = \frac{15}{150} \times 100 = 10$$

19. wt of solute $= 10gr$

wt of solvent $= 190gr; W\% = ?$

$$W\% = \frac{w}{w+W} \times 100 = \frac{10}{10+190} \times 100 = \frac{10}{200} \times 100 = 5$$

20. wt of solute $= 30gr$

wt of solution $= 200gr, w\% = ?$

$$W\% = \frac{w}{w+W} \times 100; = \frac{30}{200} \times 100 = 15$$

WORKSHEET – 3
Level – I

1. B	4. C	7. D	10. C
2. C	5. C	8. B	
3. C	6. C	9. B	

1. M = Moles / lit

2. $V_{(ml)} = 100ml, M = 0.1M$

Mwt $= 106gr$

wt = ?

$$M = \frac{Wt}{GMWt} \times \frac{1000}{V_{(ml)}}; 0.1 = \frac{x}{106} \times \frac{1000}{10}; x = \frac{0.1 \times 106}{10} = \frac{106}{100} = 1.06gr$$

3. $M = 0.05M; V_{(ml)} = 600ml$

$$M = \frac{n}{V_{(ml)}} \times 1000$$

$$0.05 = \frac{n}{600} \times 1000 \Rightarrow n = \frac{0.05 \times 6}{10} = \frac{30}{1000} = 0.03moles$$

4. $M_1 = 0.5M$

$M_2 = 0.2M$

$V_1 = 100M_1$

$V_2 = ?$

$M_1\ V_1 = M_2\ V_2 \Rightarrow 0.5 \times 100 = 0.2 \times V_2$

$V_2 = 250M_1$

Vol of H_2O must be added $= V_2 - V_1$

$= 250 - 100$

$= 150$ml

6. $M_1\ V_1 + M_2\ V_2 = M_R(V_1 + V_2); 0.1 \times 100 + 0.2 \times 200 = M_R(100 + 200)$

$10 + 20 = M_R(300); M_R = \frac{30}{300} = 0.1M$

Level – II

1. 2	9. 2	17. 3	25. 3
2. 2	10. 4	18. 1	26. 1,2,3
3. 3	11. 2	19. 2	27. 1
4. 1	12. 3	20. 4	28. 2
5. 1	13. 2	21. 3	29. 4
6. 2	14. 3	22. 2	30. 3
7. 4	15. 2	23. 1	
8. 3	16. 4	24. 3	

31. a-p;b-q;c-3;d-4

32. 1,2

33. a-p;b-q;c-r;d-p,q,s

1. Wt = 7.1gr

$V_{(ml)} = 100M_1$

MWt = 142

$M = \frac{Wt}{GMWt} \times \frac{1000}{V_{(ml)}}; = \frac{7.1}{142} \times \frac{1000}{100}; M = \frac{71}{142 \times 10} \times \frac{1000}{100} = 0.5M_2$

2. $M_1 = 0.1\left[M = \frac{N}{2} \Rightarrow N = 0.2\right]\ M_2 = 0.2[M = N]$

$N_1 = 0.2$

$N_2 = 0.2$

$V_1 = ?$

$V_2 = 40M_1$

$N_1\ V_1 = N_2\ V_2$

$0.2 \times V_1 = 0.2 \times 40; V_1 = 40$ml

3. $M_1 = 10$ m $M_2 = 0.1$ m

$V_1 = 10$ml $V_2 = ?$

$M_1\ V_1 = M_2\ V_2$

$10 \times 10 = 0.1 \times V_2$

$V_2 = \frac{100}{0.1} = 1000$ml

$\Rightarrow$ Vol of H_2O must be added $= 1000 - 10 = 990$ml (or) $= 9.9 \times 10^2$ml

4. The molarity of pure $H_2O = \frac{1000}{18} = 55.55$

5. $M = \frac{n}{V_{(ml)}} \times 100; M = \frac{0.006}{100} \times 1000; M = 0.06M$

6. $W/V = 4; M = ?,\ M = \frac{W/V\% \times 10}{GMWt}; M = \frac{4 \times 10}{40} = 1M$

7. $Wt = 0.53gr; MWt = 106gr; V_{(ml)} = 10ml$

$$M = \frac{Wt}{GMWt} \times \frac{1000}{V_{(ml)}}; M = \frac{0.53}{106} \times \frac{1000}{10}; = \frac{53}{106 \times 100} \times 100 = 0.5M$$

12. $M = \frac{n}{V_{(ml)}} \times 100; M = \frac{0.01}{500} \times 1000; = 0.02M$

13. $M = \frac{5.85}{58.5} \times \frac{1000}{500} = \frac{1}{5} = 0.2M$

14. $M_R = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$

$$= \frac{480 \times 1.5 + 520 \times 1.2}{520 + 480} = \frac{720 + 624}{1000} = \frac{1344}{1000} = 1.344M$$

15. $W/W\% = 15$

$d = 1.10\ gml^{-1}$

$M.W\% = 98$

$$M = \frac{W/W\% \times 10 \times d}{GMW}$$

$$M = \frac{15 \times 10 \times 1.10}{98}; M = 1.684M$$

16.

$$M = \frac{W/W\% \times 10 \times d}{GMW}$$

$$M = \frac{98 \times 10 \times 1.95}{98} = 19.5M$$

17.

$$M = \frac{W/W\% \times 10 \times d}{GMW}$$

$$M = \frac{98 \times 10 \times 1.84}{98} = 18.4M$$

18. $Wt = 47.5gr; V_{(ml)} = 500Ml; MWt = 95;$

$$F = \frac{Wt}{GFW} \times \frac{1000}{V_{(m1)}} = \frac{47.5}{95} \times \frac{1000}{500} = 1\ F$$

19. $Wt = 50gr$

$d = 0.936\ g/Ml$

$Wt\ of\ solvent = 500gr$

$d = 0.936\ g/M1; d = \frac{M}{V}$

$0.936 = \frac{500}{V}; V = \frac{500}{0.936} = 534.1$

$$F = \frac{Wt}{GFW} \times \frac{1000}{V_{(ml)}}; = \frac{50}{58.5} \times \frac{1000}{534.1} = 1.6\ F$$

22. $M = \frac{Wt}{GMWt} \times \frac{1000}{W_{(gr)}}; = \frac{3.42}{342} \times \frac{1000}{250} = 0.04M$

23. $M = \frac{Wt}{GMWt} \times \frac{1000}{W_{(gr)}}; \frac{3.42}{342} \times \frac{1000}{250} = 0.4M$

24. $M = 2.05M; d = 1.02\ g/M1$

$$M = \frac{Vs}{(1000 \times d) - (M \times GMW)}$$

$$m = \frac{1000 \times M}{(1000 \times 1.02) - (2.05 \times 60)}$$

$$= \frac{2050}{1020 - 123.6} = \frac{2050}{896.4} = 2.28Mol/Kg$$

25. M = 2M
d = 1.02gr/ml
MWt = 60
M = ?
$M = \frac{M \times 1000}{(1000 \times d) - (M \times GMW)}$;
$M = \frac{2 \times 1000}{(1000 \times 1.02) - (2 \times 60)} = \frac{2000}{1020 - 120} = \frac{2000}{900} = 2.22$

28. $M = \frac{d \times \text{Percent} \times 10}{\text{mol. wt}}$
$= \frac{1.1 \times 15 \times 10}{98} = \frac{150 \times 1.1}{98} = \frac{165}{98} = 1.684M$

29. $M = \frac{1.95 \times 10 \times 98}{98} = 1.95 \times 10 = 19.5$

30. $M = \frac{98 \times 10 \times 1.84}{98} = 18.4M$

WORKSHEET – 4

Level – I

1. A	4. B	7. A	10. C
2. D	5. C	8. A	
3. D	6. B	9. A	

1. Normality Units : gram eq per litre.
2. Mill ; Normal solution expressed : $\frac{N}{1000}$
3. Normality (N)
Ans: all of these
4. $N = \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2}$
5. Normality of the acid = Molarity × basicity
6. Normality × grm.eq.wt = molarity × mol. wt
7. $M = \frac{X_B \times 1000 x d}{X_A m_A + X_B m_B}$
8. N = n × molarity $\left(n = \frac{\text{Mol.wt}}{\text{eq.wt}}\right)$
9. $\frac{\text{At.Wt}}{\text{valency}}$ = Eq.wt
10. $\frac{\text{molwt.of the base}}{\text{acidity}}$ = Eq wt of base
11. wt = ?

Level – II

1. 1	8. 1	15. 3	22. 1
2. 3	9. 2	16. 4	23. 2
3. 1	10. 2	17. 1	24. 2
4. 2	11. 4	18. 4	25. 2
5. 2	12. 1	19. 4	26. 2
6. 3	13. 1	20. 1	27. 3
7. 2	14. 1	21. 4	28. 2

29. 1 30. 2
31. a-p;b-q;c-r;d-t
32. 2.68
33. 2,3
34. a-p;b-q;c-r;d-t
35. 4
36. a-t;b-p;c-q;d-r

1. $N = 0.1\ N$
$V(ml) = 250ml$
$N = \frac{w}{E.wt} \times \frac{1000}{V(ml)}$
$0.1 = \frac{x}{58.5} \times \frac{1000}{250}; x = \frac{0.1 \times 58.5}{4} = 1.46gr$

2. H_3PO_3
$HO - \underset{OH}{P} - OH$
$M = \frac{N}{2}$
$1 = \frac{N}{2} \Rightarrow N = 2\ N$

3. Wt of solute = 4gr
Wt of solution = 100ml
(V(ml)).
N = ?
$N = \frac{wt}{Ewt} \times \frac{1000}{V(ml)}; N = \frac{4}{40} \times \frac{1000}{10} = 1\ N$

4. $M = 2M; N = ?$; For sulphuric acid
$M = \frac{N}{2} \Rightarrow 2 = \frac{N}{2}; N = 4\ N$

5. $N = 0.1\ N; E.Wt = 63; V(ml) = 500ml; wt = ?$
$N = \frac{wt}{Ewt} \times \frac{1000}{V(ml)}; 0.1 = \frac{x}{63} \times \frac{1000}{500}; x = \frac{0.1 \times 63}{2}$
$= 3.15gr$

6. $M_1\ V_1 = M_2\ V_2; 18 \times 5 = M_2 \times 500;$
$M_2 = \frac{18}{100} = 0.18; N = M \times n; N = 2 \times 0.18; N = 0.36$

7. $N = 0.1\ N; V(ml) = 1500ml; N = \frac{wt}{Ewt} \times \frac{1000}{V(ml)}; 0.1 = \frac{x}{40} \times \frac{1000}{1500}$
$0.1 = \frac{x}{60} \Rightarrow x = 0.1 \times 60; x = 6gr$

8. $N = \frac{w}{Eqwt} \times \frac{1000}{V(ml)}$ $Mwt = 200, Eqwt = 100$
$0.1 = \frac{x}{100} \times \frac{1000}{100}; x = 0.1 \times 10 = 1gr$

9. Mole fraction of solute = 6 moles
Mole fraction of solution = 100 moles X_A = ?
(X_A) Mole fraction of solute = $\frac{X_A}{X_A+X_B} = \frac{6}{100} = 0.06$

10. $n_{urea} = \frac{60}{60} = 1; n_{water} = \frac{90}{18} = 5; X_A = \frac{X_A}{X_A+X_B} == \frac{1}{1+5} = \frac{1}{6}$

11. $N = \frac{wt}{Ewt} \times \frac{1000}{V(ml)}; N = \frac{6}{60} \times \frac{1000}{100} = 1\ N * 6$ gr of NaOH/100ml of H_2O

12. Molarity = 2.5; X_A = ?; Molarity Vs X_A

$$M = \frac{X_A \times 1000}{(1 - X_A)(Gmw)_B}$$

$$2.5 = \frac{X_A \times 1000}{18 - 18X_A}; 2.5(18 - 18X_A) = 1000X_A; 45 - 45X_A = 1000X_A; 45 = 1045X_A$$

$$X_A = \frac{45}{1045} = 0.043$$

14. $M = \frac{X_A \times 1000}{(1-X_A)(GMW)_B}$ $X_A = 0.15; X_B = 0.85; (GMW)_B = 18$ $M = \frac{0.15 \times 1000}{0.85 \times 18} = \frac{150}{0.15} = 9.8M$

15. $N = \frac{w}{EqWt} \times \frac{1000}{V(ml)}$; $N = \frac{w}{V(ml)} \times \frac{1000}{Eqwt}; N = \frac{10}{100} \times \frac{1000}{60}; = \frac{10}{6} = 1.66$

16. Let x lit of A and $(2 - x)$ lit of B are mixed
$M_1\ V_1 + M_2\ V_2 = M_R(V_1 + V_2); 0.5 \times x + 0.1(2 - x) = 0.2 \times 2$
$0.5x + 0.2 - 0.1x = 0.4; (0.5 - 0.1)x = 0.4 - 0.2; 0.4x = 0.2; x = 0.5;$ A is x = 0.5 and B is $x = 1.5$

18. d = 0.6 g/ml; w/W% = 34%; GEW = 35

$$N = \frac{10 \times d \times w/W\%}{GEW}; N = \frac{10 \times 0.6 \times 34}{35}; N = 5.8\ N$$

19. $N = \frac{W}{Eqwt} \times \frac{1000}{V(ml)}; N = \frac{6.3}{63} \times \frac{1000}{250} = 0.4; N_1\ V_1 = N_2\ V_2; 0.4 \times 250 = 0.1 \times V_2; V_2 = 1000$
Vol of H_2O to be added = $V_2 - V_1 = 1000 - 250 = 750$ml (or) 0.75ml

20. NVsw/V%; $N = \frac{10 \times w/V\%}{greq\ wt}$

21. $N = \frac{wt}{gr Eqwt} \times \frac{1000}{V(ml)}; 0.05 = \frac{x}{49.04} \times \frac{1000}{100}; x = \frac{0.05 \times 49.04}{10}; x = \frac{0.05 \times 49.04}{10} = 0.2452$

22. $n_{H_2O} = \frac{36}{18} = 2,\ n_{CH_3CHO} = \frac{64}{44} = 1.45,\ X_B = \frac{n_A}{n_A+n_B} \Rightarrow X_B = \frac{1.45}{2+1.45} = \frac{1.45}{3.45} = 0.42$

23. $N = \frac{40}{20} \times \frac{1000}{1000}$ $N = 2\ N$

24. No of millimoles = $N \times V(ml); 10 = N \times 50; N = \frac{1}{5} = 0.2\ N$

25. $N = 0.2\ N; V(ml) = 250ml$

$$N = \frac{\text{No gr eq}}{V_{(ml)}} \times 1000$$

$$0.2 = \frac{x}{250} \times 1000; x = \frac{0.2}{4} = 0.05\ N$$

26. $\frac{8}{56} \times \frac{1000}{100} = 1.42\ N$

27. $\frac{6}{40} \times \frac{1000}{100} = 1.5\ N$

28. 1 N

29. $\frac{24.5}{98} \times \frac{1}{1} = 0.25\ N$